

Task #2

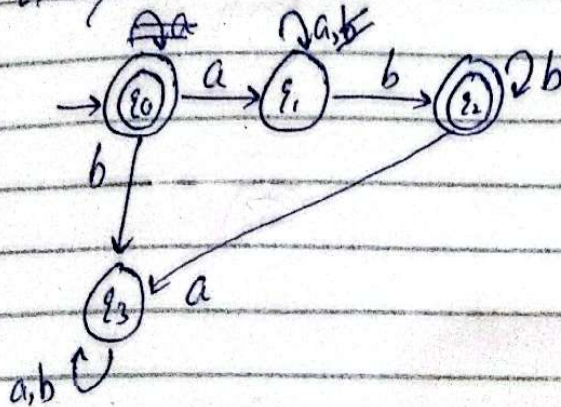
Roll no:- 24K-0737

Name:- Yash Raj

⇒ Construct DFA's:-

$L_1: \{a^n b^m : (n+m) \text{ is even}\}$

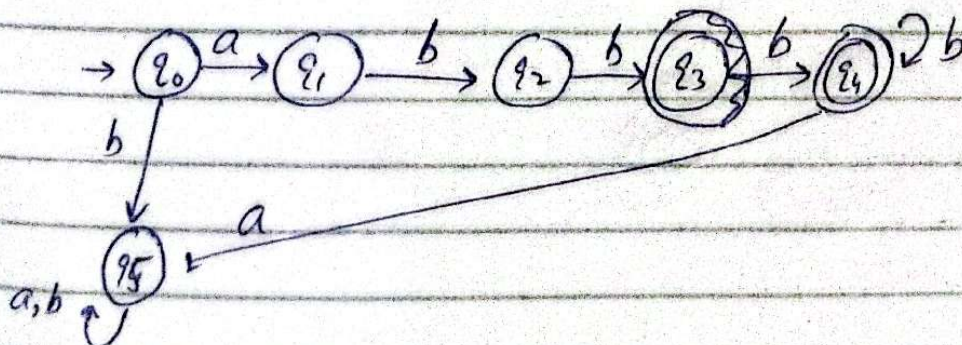
$L_1: \{\lambda, ab, aabb, aaabbb, \dots\}$



$aaab.$
 $aaabbb$
 $aaaaabbb$

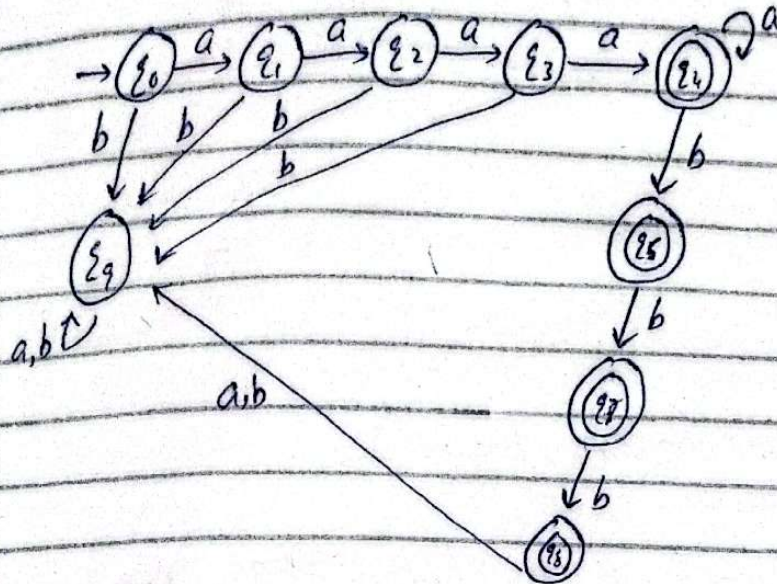
$L_2: \{ab^n \mid n \geq 3, w \in \{a,b\}^+\}$

$L_2: \{abbb, abbbb, abbbbbb, \dots\}$



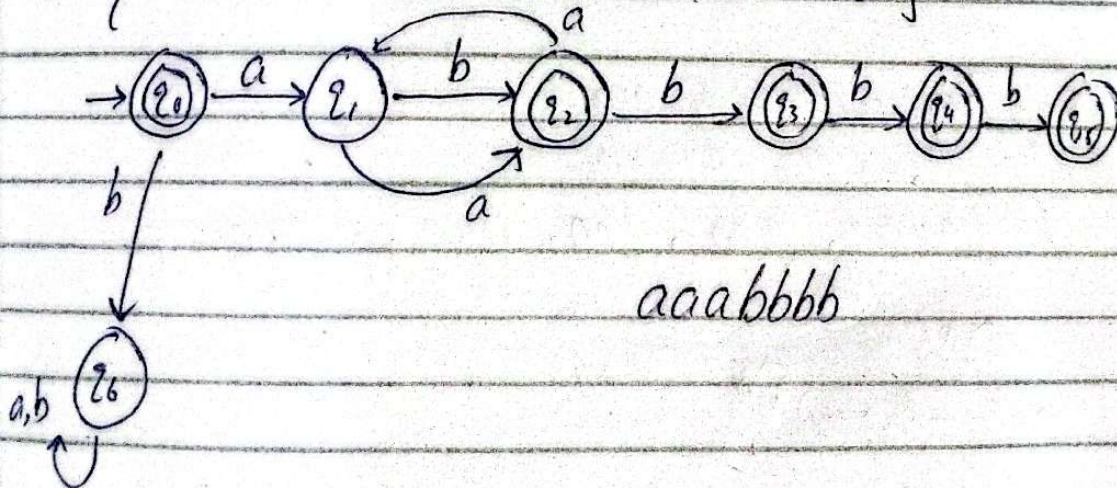
$L_2 = \{a^n b^m, n \geq 4, m \leq 3\}$

$L_2 = \{aaaa, aaaaab, aaaaabb, aaaaabbb, aaaaabbbb \dots\}$



$L_3 = \{a^n b^m, n \leq 4, m \leq 4\}$

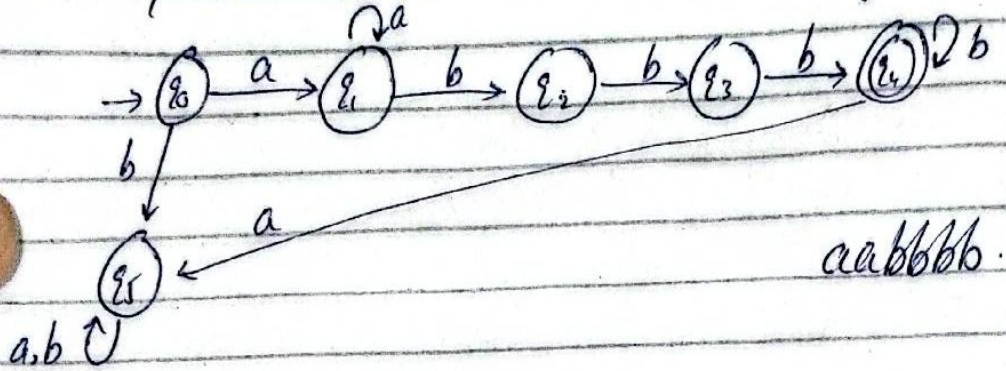
$L_3 = \{\epsilon, ab, aabb, aaabbb, aaaabbbb\}$



aaaabbbb

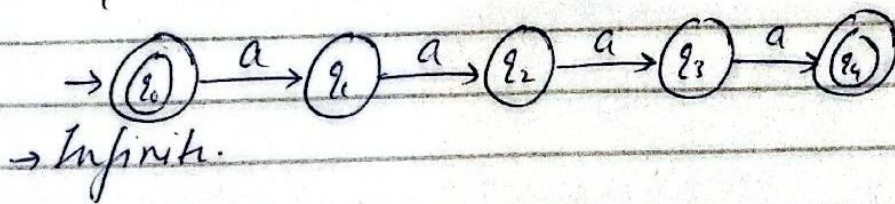
$L_4: \{a^n b^m : n \geq 1, m \geq 1, nm \geq 3\}$

$L_4: \{abbb, aaab, aabb, aabbb, \dots\}$



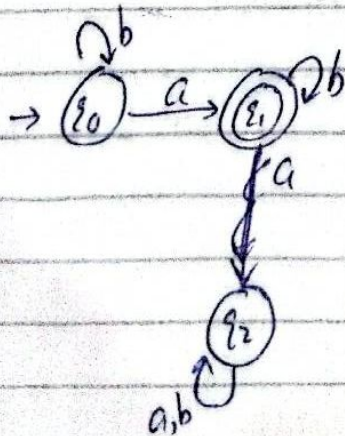
$L_6: \{vwv : v, w \in \{a,b\}^+, |v| \geq 2\}$

$L_6: \{aa, aaaa, abab, baba, bbbb, aabaa, abbbab, \dots\}$



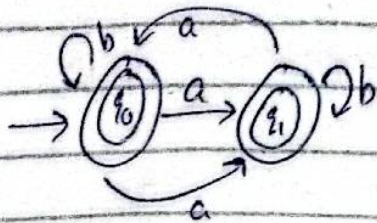
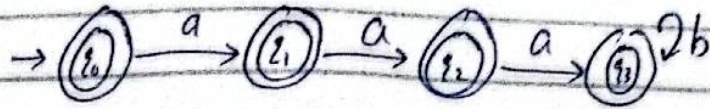
$L_8: \text{having exactly one } a$

$L_8: \{a, ab, abb, abbb, bab, babb, \dots\}$



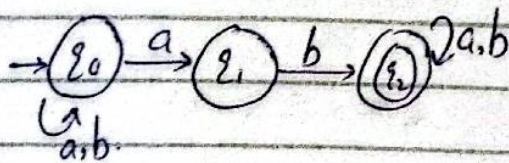
L_9 , string containing no more than 3 a's

L_9 , $\{ \epsilon, a, aa, aaa, aaab, baaab, bababa \dots \}$



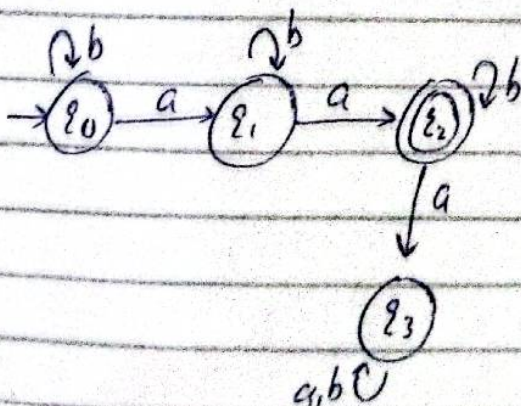
L_{10} , at once occurrence of each symbol in Alphabet.

L_{10} , $\{ ab, abb, bab, bbaabb \dots \}$



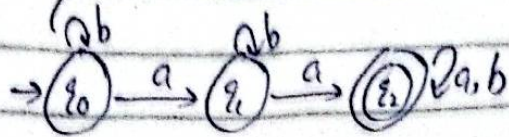
L_{18} , exactly two a's

L_{18} , $\{ aa, aab, baab, baba \dots \}$



L_{19} : at least two a's:

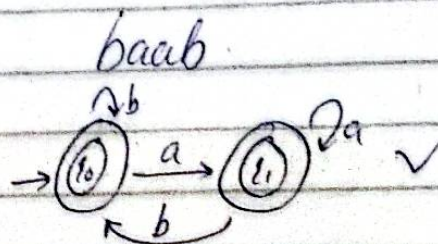
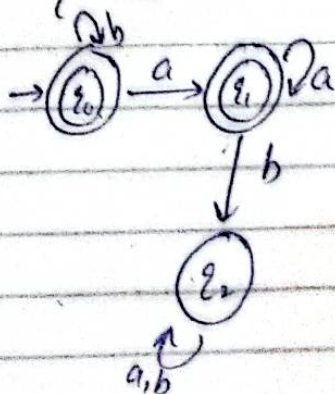
$L_{19} = \{aa, aaa, aaaaab, baab, \dots\}$



baba.
aababa.

L_{20} : all strings that don't end with ab.

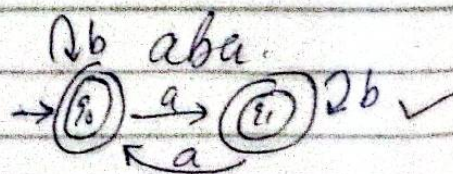
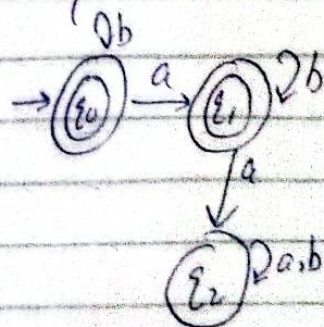
$L_{20} = \{\lambda, a, b, baa, baba, aabaa, \dots\}$



baab.
bbbab. x
babba. ✓
ababaa. ✓
baa. ✓

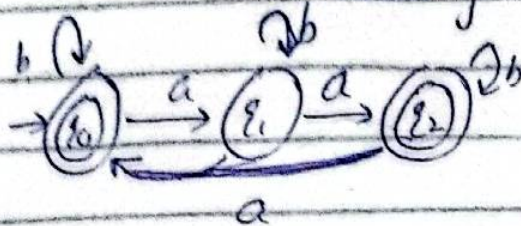
L_{22} : strings not containing the substring aa.

$L_{22} = \{\lambda, a, ab, baba, abba, \dots\}$



L_{22} : language of all strings in which the number of a 's is even.

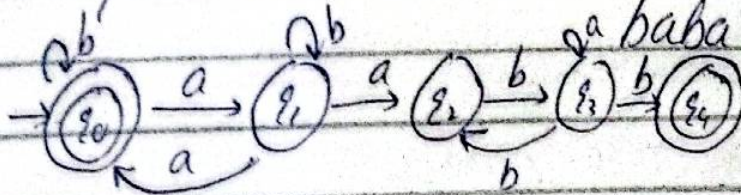
L_{23} : $\{ \lambda, aa, aaaaab, baba, aababa, aab \}$.



babababa.

L_{24} : number of a 's & b 's both even.

L_{24} : $\{ \lambda, aabb, aaaaabbbb, aaaaaabbbbbbb -- \}$.



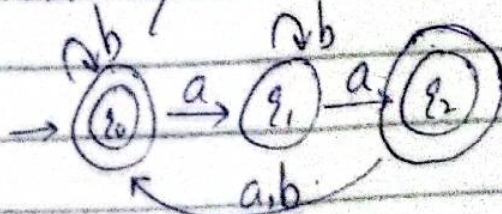
aaaaabbbb.

baab.

aaaaaabbbbbbb.

L_{25} : no more than one occurrence of aa .

L_{25} : $\{ \lambda, aa, aab, abaa, \dots, baab \}$.

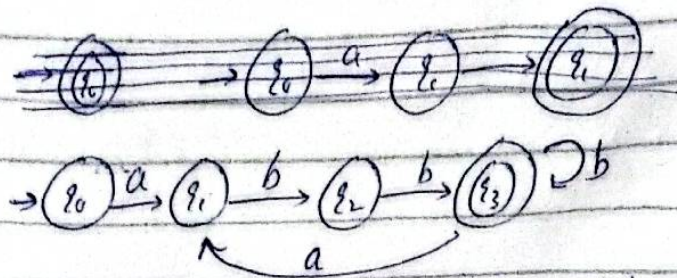


abababa.

baaab.

L_{26} : every a is followed immediately by bb

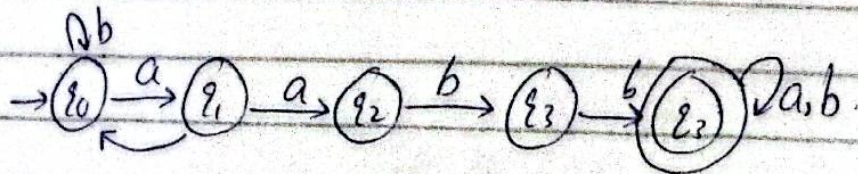
$L_{26} = \{abb, babb, bbabbabb, \dots\}$



$abbabbabb$

L_{27} : language of all strings containing both bb & aa as (sub)strings

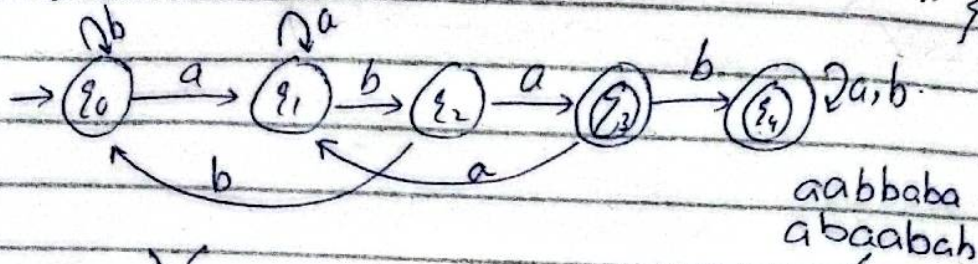
$L_{27} = \{aabb, baabb, bbaa, \dots\}$



$abbaab$
 $aabbba$
 $bbaa$
 $bbbaab$
 $baabbba$
 $aabbba$

L_1 : all strings containing both aba & bab as substring.

$L_1 = \{ abab, aabab, \text{babab}, bababb, \dots \}$



$q \backslash \Sigma$	a	b
Σ		
q_0	q_1	q_0
a q_1	q_1	q_2
ab q_2	q_3	q_0
aba q_3	q_1	q_4
abab q_4	q_3	q_4

~~q_0~~ , ~~q_1~~ , ~~q_2~~

~~aa~~ ~~ab~~
~~aba~~ ~~abb~~

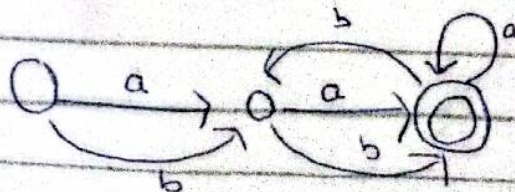
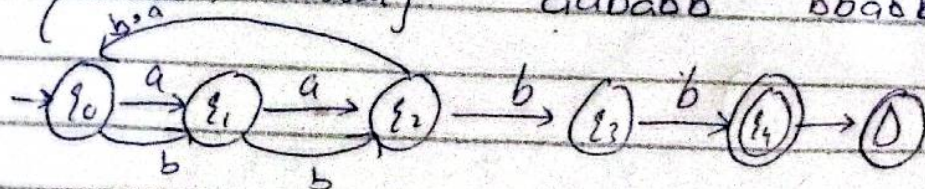
~~abba~~ ~~abab~~

~~ababa~~ ~~ababb~~

L_2 : The language of all strings that begin or end with aa or bb.

$L_2 = \{ aabb, bbaa \}$

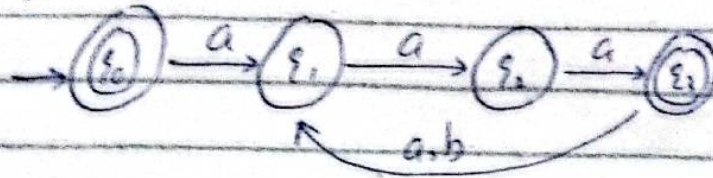
aababb bbabba



aababb
bbabba

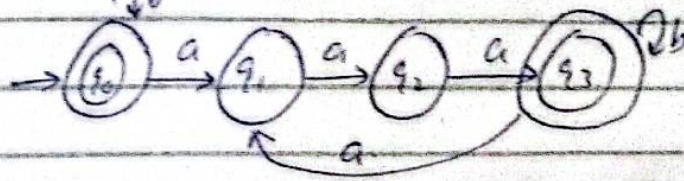
$$L_6 = \{w: |w| \bmod 3 = 0\}$$

$$L_6 = \{\epsilon, aaa, bbb, aab, baa, \dots\}$$



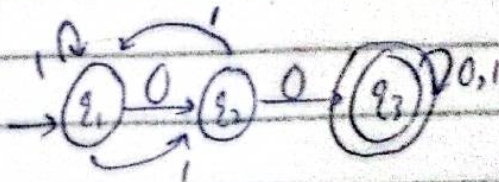
$$L_7 = \{w: n_a(w) \bmod 3 = 0\}$$

$$L_7 = \{\epsilon, aaab, aaaaaab, aaaaaabb, baaaaaa, \dots\}$$



$L_8 =$ having ^{exactly} one pair of consecutive zeros

$$L_8 = \{00, 001, 100, 10100, 00101\}$$



1001010

1001

01001

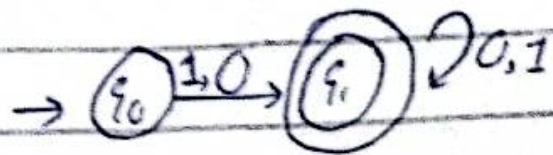
11001, 0110100

11001

101001

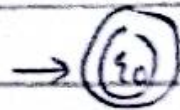
$L_{11} = \{\text{all strings ending in } 0, 1\}$

$L_{11} = \{0, 1, 101, 100\}$

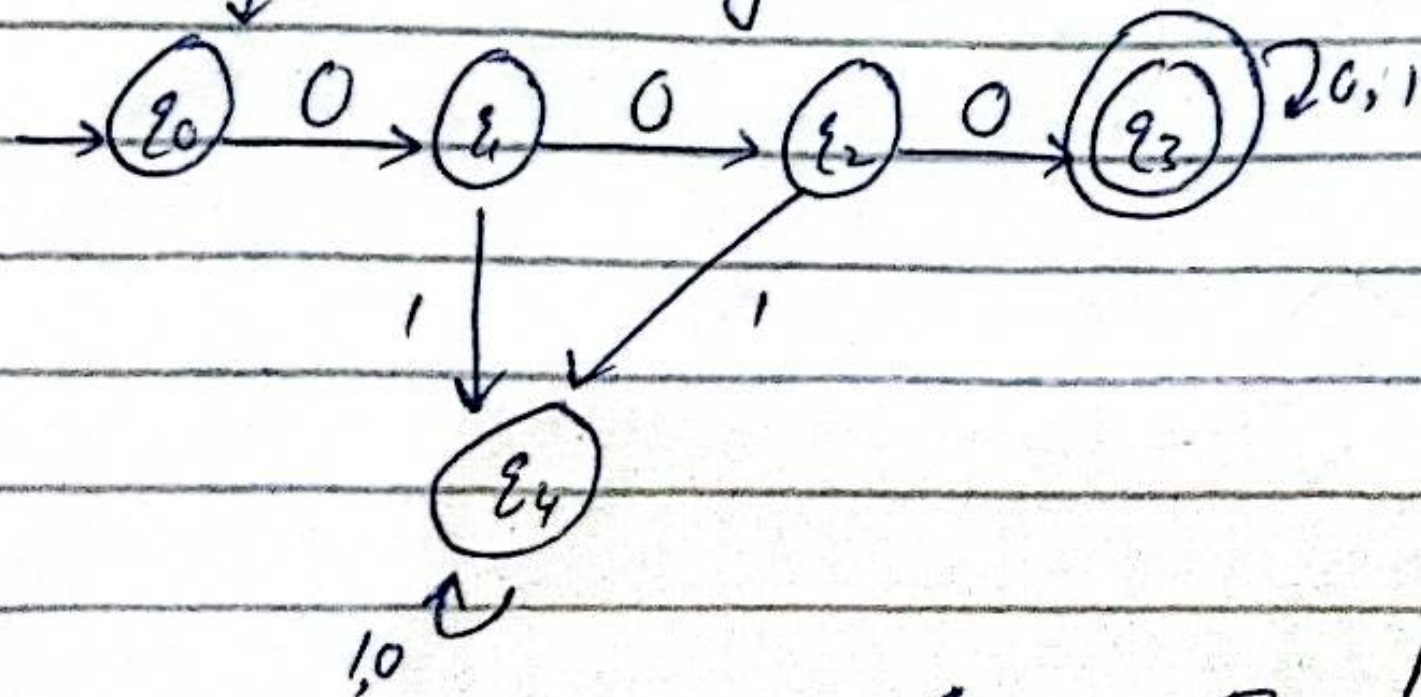


$L_{12} = \{\text{all strings not ending in } 0, 1\}$

$L_{12} = \{\lambda\}$



$L_14 = \{ \text{all strings having at least two occurrences of sub string } 00 \}$



100001
0000
000
100001

$L_1 = \{ \text{not containing } 101 \}$

$L_1 = \{ \lambda, 0, 1, 001, \dots \}$

